

LO5 -- Container & Kubernetes Troubleshooting Guide (Expanded)

Universal Troubleshooting Workflow

1. Check object state

```
kubectl get pods
```

2. Inspect events

```
kubectl describe pod POD
```

3. Read logs

```
kubectl logs POD
```

```
kubectl logs POD --previous
```

4. Enter the container

```
kubectl exec -it POD -- sh
```

5. Verify Service

```
kubectl get svc
```

```
kubectl get endpoints
```

6. Check events

```
kubectl get events --sort-by=.lastTimestamp
```

Common Errors

Pending

Possible causes: - Insufficient CPU - Insufficient Memory - Missing PVC - nodeSelector mismatch - Taints

Diagnosis:

kubectl describe pod

ImagePullBackOff

Causes: - Typo - Missing image tag - Private registry - Missing imagePullSecret

Fix: Correct image or configure registry secret.

CrashLoopBackOff

Meaning: Application starts then crashes repeatedly.

Diagnosis:

kubectl logs --previous POD

Typical causes: - Missing configuration - Wrong command - Database unavailable

OOMKilled

Container exceeded memory limit.

Check:

kubectl describe pod

Fix: Increase limits or optimize application.

FailedMount

PVC not found or cannot bind.

Verify:

kubectl get pvc

kubectl describe pvc

Service not working

Check in order:

1. Pod running
2. Labels
3. Service selector
4. Endpoints
5. DNS
6. TargetPort

Readiness Probe Failure

Symptoms: - Pod Running - READY = 0/1

Fix readiness probe or application.

Liveness Probe Failure

Symptoms: Container repeatedly restarts.

Fix application or probe settings.

Podman Problems

Wrong port mapping:

-p HOST:CONTAINER

Host networking:

--network host

ignores -p.

Busybox exits immediately because it has no long-running process.

Fix:

sleep infinity

SELinux bind mount:

-v ./html:/usr/share/nginx/html:Z

Containers cannot resolve names on different networks.

Dockerfile Best Practices

- Combine `apt-get update && apt-get install`.
- Copy dependency files before application code.
- Use exec-form CMD:

CMD ["python","app.py"]

- Use multi-stage builds.
 - Run as non-root.
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YAML Problems

Common mistakes: - Wrong indentation - Wrong apiVersion - Wrong kind - Selector mismatch - Typo in imagePullPolicy

Validate:

kubectl apply --dry-run=server -f file.yaml

OpenShift vs Kubernetes

Route: - Native OpenShift - Built-in TLS

Ingress: - Kubernetes standard - Requires Ingress Controller

High-Value Commands

kubectl describe

kubectl logs

kubectl exec

kubectl get events

kubectl get endpoints

kubectl rollout history

kubectl rollout undo

kubectl get all

podman logs

podman inspect

podman stats

Exam Strategy

Always explain: - What failed - How you diagnosed it - Which command you used - Why the fix solves the issue